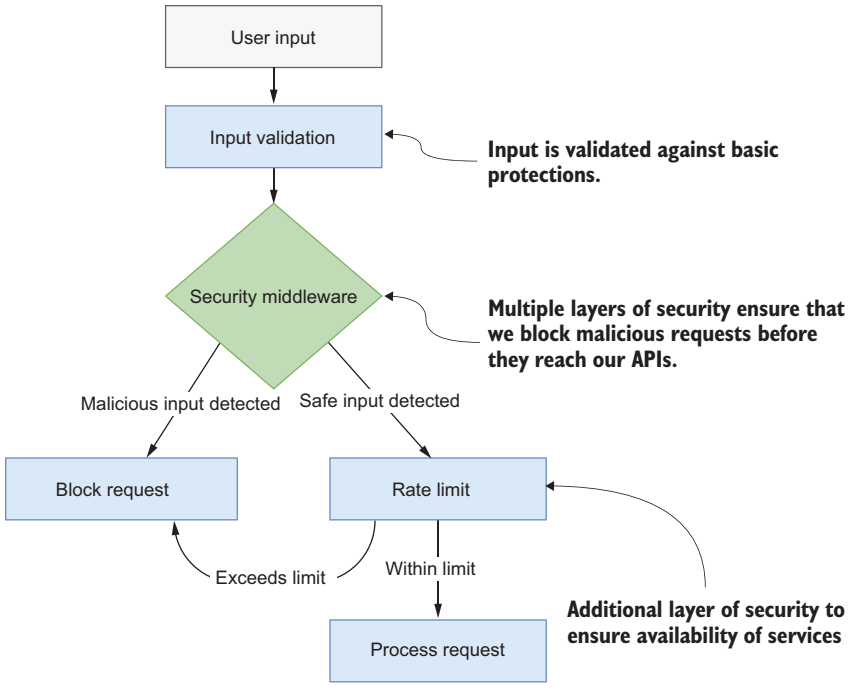


Figure 9.1 The multilayered security checks from user input validation through request processing, showcasing multiple defensive layers including input validation, security middleware, and rate-limiting mechanisms

Generally, security should follow a multilayered approach like the one shown in figure 9.1, where each request flows through progressively more sophisticated security checks. Only requests that successfully navigate all these security layers proceed to final processing, creating a robust defense system that both protects our AI resources and maintains service quality for legitimate users.

9.1.1 Input validation

Input validation ensures that user inputs conform to expected formats and constraints. With input validation, we can significantly reduce the risk of processing malicious data that could lead to security vulnerabilities, such as prompt injection attacks or excessive data usage. Before we even consider what to validate, we need to do some work to establish our web application's threat model.

ESTABLISHING A THREAT MODEL

A threat model is a process to help identify potential vulnerabilities and assess which components of the application are most at risk. This process involves analyzing various aspects of an application, including